

DATA 642

Quiz 4

1. What practical skills are developed in implementing Kernel PCA?
 - A) Feature extraction from unstructured data
 - B) Data preprocessing for regression tasks
 - C) Dimensionality reduction using kernel functions
 - D) Model interpretation for complex datasets
2. How can PCA be applied for feature selection?
 - A) By selecting features with the highest variance
 - B) By selecting features with the lowest variance
 - C) By selecting features with the highest covariance
 - D) By selecting features with the lowest covariance
3. What is the primary purpose of using PCA as a preprocessing step for machine learning models?
 - A) To increase the complexity of the model
 - B) To decrease the interpretability of the model
 - C) To enhance model efficiency and reduce computational complexity
 - D) To introduce non-linearity into the model
4. What is the significance of statistical independence in data analysis?
 - A) It ensures that all variables have equal importance
 - B) It simplifies the modeling process by reducing complexity
 - C) It allows for the identification of underlying patterns in data
 - D) It increases the computational cost of analysis
5. What are the assumptions underlying Independent Component Analysis (ICA)?
 - A) Gaussian distribution of data
 - B) Statistical independence between underlying variables

- C) Linear relationships between variables
 - D) Equal variances among variables
6. What is the role of the non-negativity constraint in matrix factorization?
- A) To ensure that all matrix elements are positive
 - B) To enforce sparsity in the matrix representation
 - C) To prevent overfitting in the model
 - D) To reduce the computational complexity of factorization
7. How does Non-negative Matrix Factorization (NMF) promote sparse and interpretable representations?
- A) By decomposing the matrix into orthogonal factors
 - B) By minimizing the reconstruction error of the original matrix
 - C) By enforcing non-negativity constraints on the factors
 - D) By maximizing the mutual information between variables
8. What is the primary objective of Canonical Correlation Analysis (CCA)?
- A) To identify latent variables in the data
 - B) To reduce the dimensionality of the data
 - C) To find correlated patterns in different datasets
 - D) To perform clustering analysis on the data
9. What is the connection between tensorial decompositions and machine learning applications?
- A) Tensorial decompositions are primarily used for image processing tasks
 - B) Tensorial decompositions are utilized in natural language processing algorithms
 - C) Tensorial decompositions enable the extraction of latent features from multi-dimensional data
 - D) Tensorial decompositions are limited to theoretical mathematics and have no practical applications
10. Consider a $2 \times 2 \times 2$ tensor $\mathcal{X}$ with frontal slices:

$$\mathbf{X}_1 = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{X}_2 = \begin{bmatrix} 2 & 2 \\ -1 & -1 \end{bmatrix}$$

- (a) Compute the mode-3 unfolding $\mathbf{X}_{(3)}$ of this tensor.
- (b) Suppose we perform a Tucker decomposition where the core tensor $\mathcal{G}$ is of size $2 \times 2 \times 1$. Using the HOSVD procedure, compute the mode-3 factor matrix $\mathbf{C}$ for this decomposition.

You can use this page to answer the last question!